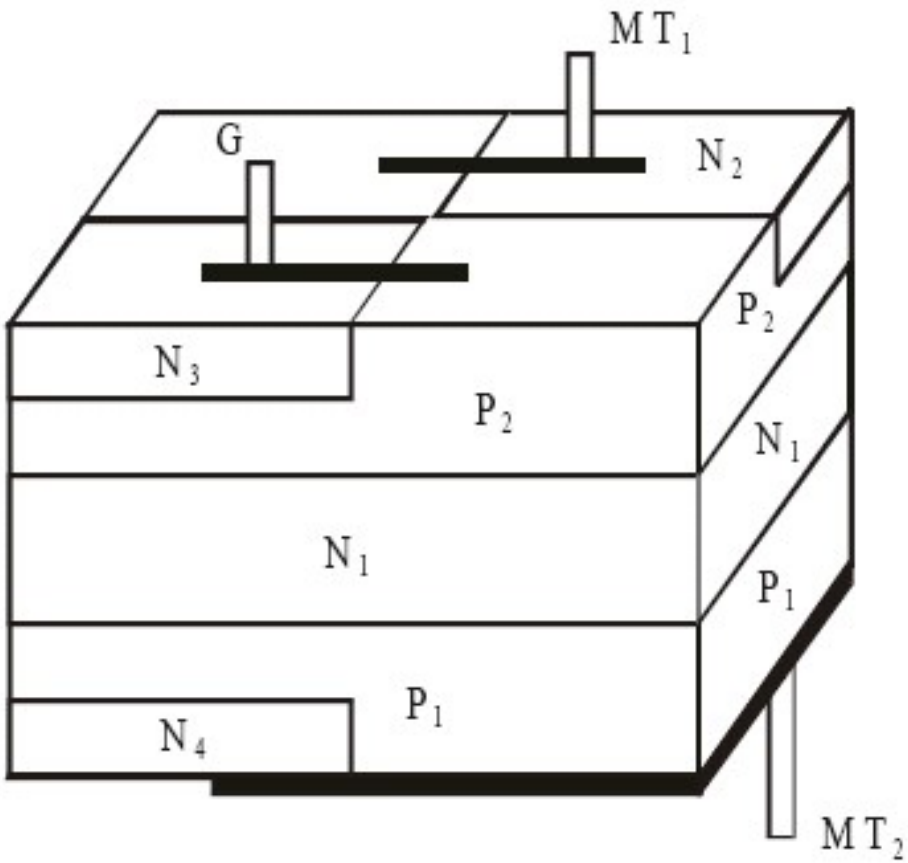
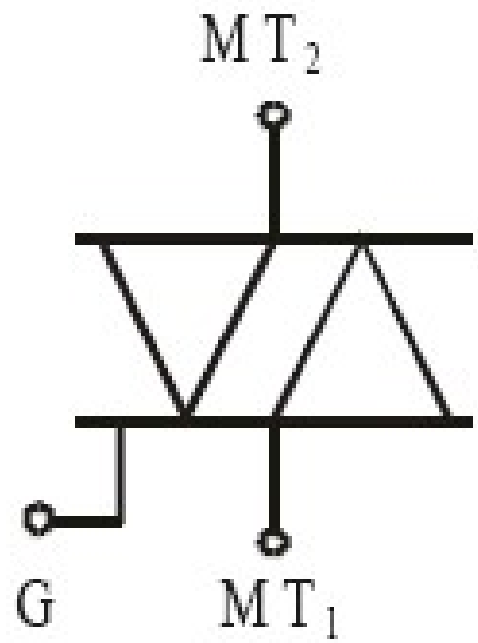
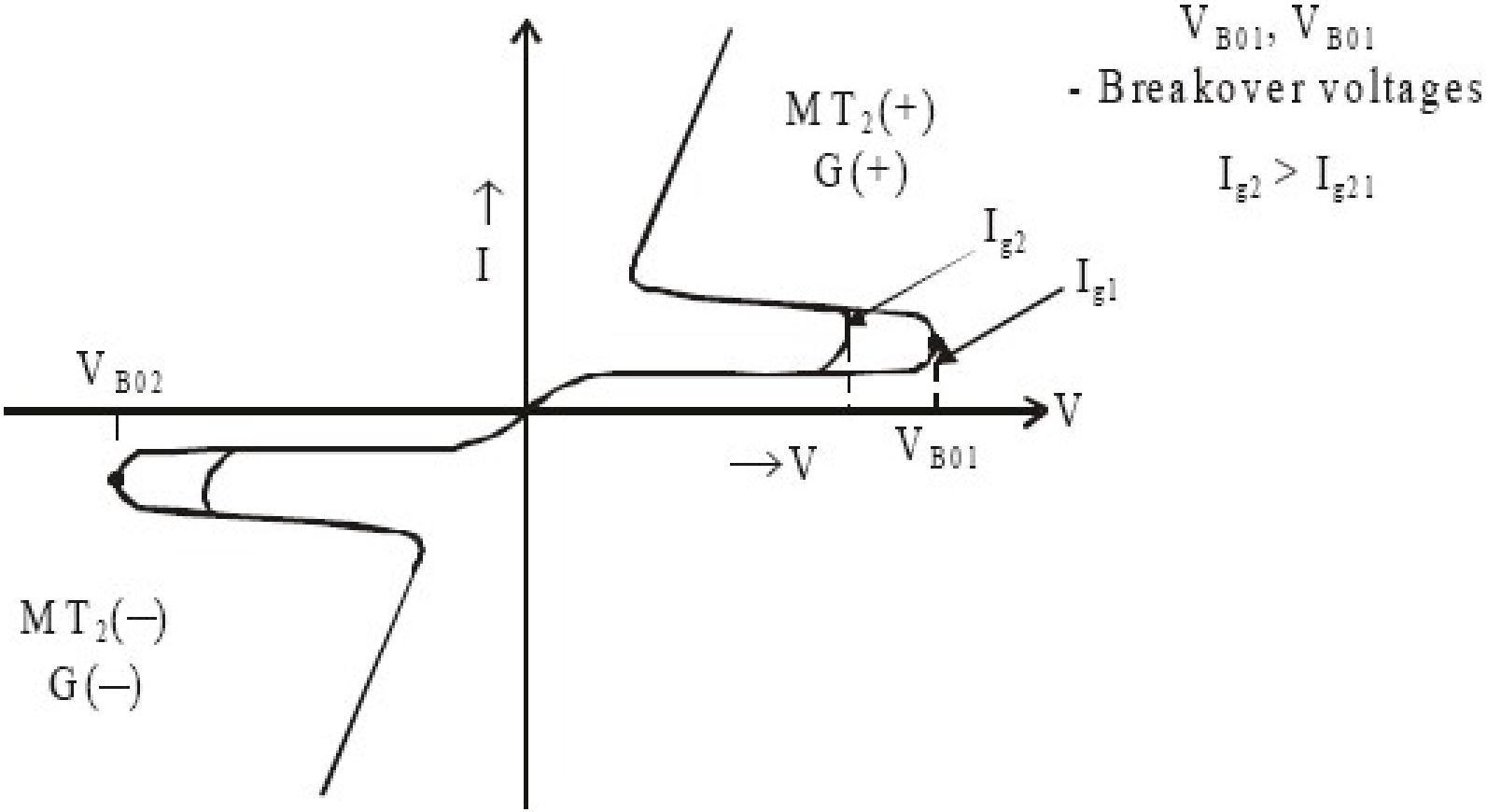
Bidirectional Triode Thyristors (TRIAC)





Triac Characteristics



Gate Turn-off Thyristors

- Turned on by applying positive gate signal.
- Turned off by applying negative gate signal.
- On state voltage is 3.4V for 550A, 1200V GTO.
- Controllable peak on-state current I_{TGQ} is the peak value of on-state current which can be turned-off by gate control.

Advantages over SCRs

- Elimination of commutating components.
- Reduction in acoustic & electromagnetic noise due to elimination of chokes.
- Faster turn-off, therefore can be used for higher switching frequencies.
- Improved efficiency of converters.

Advantages over BJTs

- Higher voltage blocking capabilities.
- High on-state gain.
- High ratio of peak surge current to average current.
- A pulsed gate signal of short duration only is required.

Disadvantages of GTOs

- On-state voltage drop is more.
- Due to multi cathode structure higher gate current is required.
- Gate drive circuit losses are more.
- Reverse blocking capability is less than its forward blocking capability.